KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2016
Department of Computer Science and Engineering
CSE 4127
Computer Vision and Image Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer vision? How is image processing related to computer vision? Mention some challenges of computer vision problem. (10)

b) Define histogram equalization. Consider the 5×5, 3 bits/pixel image shown in Fig. 1(b). (18)

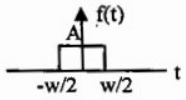
$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 4 & 5 & 3 \\ 4 & 3 & 2 & 6 & 7 \end{bmatrix}$$

Fig. 1(b)

i) Perform histogram equalization to the image.
ii) What will happen if we apply the same process on the resultant image of step (i).
iii) Explain why the discrete histogram equalization does not, in general, yield a flat histogram.

c) Perform Fourier transform of the function shown in the following figure follows from (07)

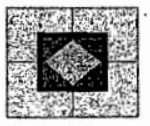
$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t}dt$$



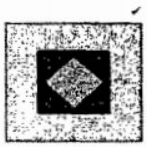
2. a) Prove that the convolution of an image in spatial domain is equal to multiplication in frequency domain. (13)

b) What does it mean by low pass filter? Design an Ideal Low-Pass Filter (ILPF). Explain why there is a ringing effect on the image that is smoothed by ILPF. (10)

c) Consider the following figure, when the image on the left was filtered using a smothing filter, the result was image on the right. The filter used was one of these of size 3×3: (i) mean filter; (ii) ideal low-pass filter; (iii) Gaussian low-pass filter; (iv) median filter. For each of the four possible filters listed above, give at least one reason why you think it was or was not, the filter actually used. (12)



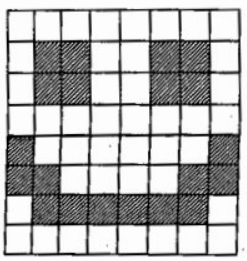
original image



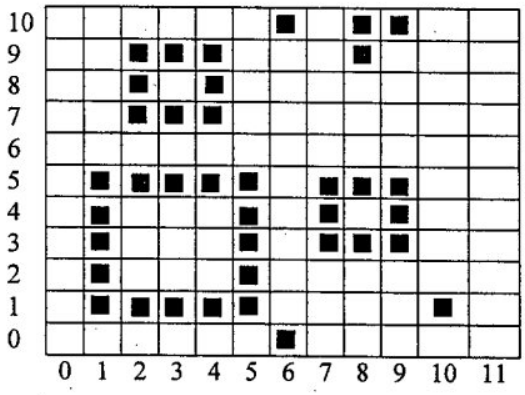
filtered image

3. a) Sketch and explain a model for image restoration process. How can you estimate the image degradation functions? (12)

b) What is the importance of image segmentation? Segment the binary image shown in following figure. Using split-and-merge technique. Also, show the quad tree corresponding to the segmentation. (12)



c) If we apply the Hough transform on the following image, what would be the maximum value for the accumulator cell in the (P,Q) space? What is the corresponding (P,Q) value? (11)



4. a) Explain how the k-means clustering can be used for image segmentation. What are the weaknesses of k-means clustering algorithm? How can they be overcome? (13)
   b) What is motion field? Explain a direct method for estimation of motion field stating the key assumptions and deriving the necessary equations. (15)
   c) State the steps of calculating Ostu's threshold for an image. (07)

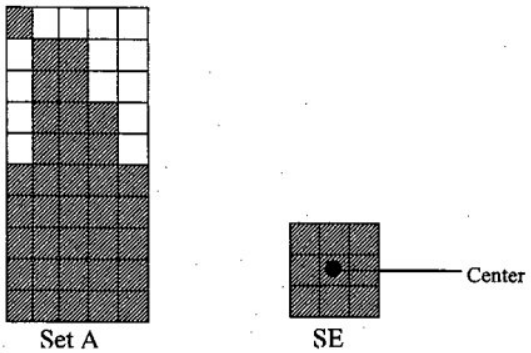
SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image processing? What are the fundamental steps in digital image processing? (09)
   b) Discuss about different neighborhood and connectivity of a pixel with appropriate example. (12)
   c) Define blind spot and brightness adaption. State some example of optical illusions. (06)
   d) What do you mean by image compression? Briefly describe any one of the image compression techniques. (08)

6. a) What is discretization? How are images presented in computer? (07)
   b) What is the property of an adaptive filter? Write a pseudocode for adaptive median filter. Instead of median filter, when will you prefer the adaptive one? Explain why. (12)
   c) State and explain the data redundancies that can be identified in an image and can be exploited for image compression. (10)
   d) Explain how human visual system perceives a color. (06)

7. a) Define convex hull. Write down the drawbacks of convex hull and how this can be resolved? (07)
   b) Define Thinning. Draw the skeleton of the following binary image. Is it possible to recover the original image from the skeleton? (15)



   c) Why pruning is necessary? Write down the equations to compute pruning of a binary image. (06)
   d) Define Erosion and Dilation for Gray-scale image with appropriate example. (07)

8. a) When we use top-hat and bottom-hat transformation in grayscale image? (04)
   b) Define Hue, saturation and intensity. Write down the process of conversion of a pixel value from RGB space to HSI space. (18)
   c) Define pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   d) How can you apply a spatial filter on a full color image? Explain. (05)